GLOBAL MACRO INTELLIGENCE BRIEF

Under-Covered Structural Risks in the World Economy

Set 3 · Chokepoints: the physical limits under the digital economy

Seven deep dives · Threat meters, plain-language summaries & reader actions

Prepared: 25 June 2026

The theme of this set. Sets 1 and 2 were about the financial system — hidden leverage, and the habit of postponing losses. Set 3 looks underneath all of it, at the **physical and structural chokepoints** the whole economy quietly runs through: electricity, copper, water, food and fertilizer, the single machine that makes advanced chips, the narrow sea lanes and undersea cables that carry trade and data, and the handful of financial hubs that clear it all. The digital and AI booms are sold as weightless. They are not. They rest on a physical substrate that is finite, concentrated, and — increasingly — contested. When a chokepoint binds, the abstract economy feels it fast.

Format (same as Set 2): each story has a 1–10 threat meter, a “Who this hits” tag, both-sides analysis, a stress-test, a plain-language summary, realistic reader actions, and forward indicators. Full sources at the end. Informational analysis only — not investment, legal, or financial advice.

How to read this brief

Each deep dive runs in this order:

- **Who this hits** — instantly see whether it touches you.
- **Current threat level (1–10)** — with what would raise or lower it.
- **What is happening / Why under-covered / The debate / Stress test / Historical parallel** — the verified analysis.
- **In plain terms** (blue box) and **What you can do now** (green box).
- **What to watch** — the forward indicators.

Confidence tags: **[Established]** multiple strong sources agree. **[Contested]** credible sources disagree. **[Single-source]** one source — a lead, not a fact. **[Forecast]** a projection, not an outcome.

Threat-level bands: **1–2 Low**, **3–4 Moderate**, **5–6 Elevated**, **7–8 High**, **9–10 Severe**. A judgement of the chance of a materially disruptive event in roughly the next 12 months, weighted by how far the damage would spread — not a forecast.

The connective thesis. Read together, these seven describe one thing: the booming, abstract top of the economy (AI, cloud, markets) is bolted onto a narrow, physical, under-invested base. Each chokepoint is a place where decades of treating a vital input as free and infinite are colliding with the reality that it is neither. The closing section maps how they interlock — and links them back to Sets 1 and 2.

Risk dashboard — Set 3 at a glance

Current threat levels across all seven stories, as of 25 June 2026. See each story for the reasoning and the indicators that would move it.

Story	Level	Band	Key driver of the level
1. AI power & the grid bottleneck	6/10	Elevated	The binding constraint on AI now
2. Copper & the electrification deficit	4/10	Moderate	Structural, but mostly post-2029
3. Water — the underpriced input	4/10	Moderate	Slow, but regionally acute
4. Food & fertilizer fragility	6/10	Elevated	Active export curbs + Hormuz inputs
5. The lithography (ASML) chokepoint	5/10	Elevated	Extreme single point of failure
6. Sea lanes & undersea cables	7/10	High	Live energy-and-trade crisis
7. Clearinghouses (financial chokepoint)	4/10	Moderate	Concentrating, no near-term trigger

Aggregate Set 3 level: **≈5/10 — Elevated** (mean ≈ 5.1). One story (sea lanes) is in active crisis; the rest are slow-burning physical constraints that the digital boom has been able to ignore — until now.

Contents

1. AI’s power problem: the grid is the new bottleneck

WHO THIS HITS: *Every household with an electricity bill — and every investor exposed to AI, because power, not chips, is now the limit on the whole buildout.*

CURRENT THREAT LEVEL: **6/10 — Elevated** *The binding constraint on the AI boom*

1	2	3	4	5	6	7	8	9	10

Why this level: Power has become the single biggest constraint on the AI buildout; household electricity bills are rising and turning political; and grid and transformer lead times are measured in years. **Raises it:** A heatwave-driven grid emergency tied to data-center load; sharper residential bill spikes feeding backlash and moratoria. **Lowers it:** Ratepayer-protection rules holding; on-site and behind-the-meter power scaling; a genuine demand slowdown (which early data hint at).

What is happening

[Established] The constraint on artificial intelligence has shifted from chips to electricity. NVIDIA’s CEO put it bluntly: every future data center will be “power-limited.” US data-center electricity demand has jumped from around 23 GW in 2023 toward 42 GW in 2026, and the wait to connect a new power source to the grid has more than doubled — from about 22 months in 2000 to roughly 54 months now. Building a data center takes about two years; new transmission can take a decade. The mismatch is the story.

[Established] Households are paying for it. US residential electricity prices rose about 42% over five years — faster than overall inflation — and far more in data-center hotspots (Washington, D.C. up ~94%, Maryland ~74%, over five years to March 2026). In the PJM grid region, capacity prices rose a staggering ~833% between the 2024–25 and 2025–26 delivery years. The Dallas Fed estimates wholesale power prices could rise up to 50% as data-center demand doubles. It has become a political issue: at least 11 US states are weighing temporary bans on new data centers, and in March 2026 seven AI companies signed a White House-brokered “Ratepayer Protection Pledge” to fund their own grid upgrades.

[Contested] How binding is the limit? The IEA estimates one-fifth of global data-center investment is at risk of delay from grid bottlenecks, and Morgan Stanley sees US demand reaching 74 GW by 2028 against a ~49 GW shortfall. But Wood Mackenzie reported the buildout actually *slowing* in late 2025 (new pipeline capacity halved quarter-on-quarter), and analysts warn of “phantom” interconnection requests inflating the forecasts.

Why it is under-covered

“AI is booming” is the headline; “the transformer has a four-year lead time” is not. The bottleneck lives in substations, interconnection queues, and utility rate cases — unglamorous, technical, and slow — even though it may decide whether the AI capex in Set 2 actually earns a return.

The debate

The shortage camp argues: demand is real and exploding, the grid physically cannot keep up, and power scarcity will throttle AI, raise everyone’s bills, and hand an edge to China (which already generates more than twice as much electricity as the US).

The skeptic camp argues: forecasts are inflated by speculative and double-counted interconnection requests; efficiency gains and a capex slowdown are already visible; and much of the “crisis” is utilities and generators talking their book into higher returns.

Both sides accept: grid and transformer lead times are long, residential bills are rising, and data centers are now a first-order driver of US electricity demand for the first time in decades.

Stress test (the benign case)

If hyperscalers fund their own power (on-site gas, nuclear, behind-the-meter) and ratepayer-protection rules hold, households can be largely shielded and the buildout proceeds without a grid crisis. A demand slowdown — already hinted at — would relieve pressure further. The weakness in that case: power infrastructure is lumpy and slow, the political backlash is already here, and a single hot summer with data centers competing against air-conditioning load is the kind of event that turns a slow problem into a sudden one.

Historical parallel

The closest analogue is the late-1990s–early-2000s telecom and dot-com infrastructure rush, when the physical buildout (fiber, then power) lagged or overshot the hype. More apt may be past episodes of demand outrunning grid investment — California's 2000–01 electricity crisis showed how quickly a supply-demand mismatch, plus market design flaws, can produce price spikes and blackouts. The lesson: electricity is not a commodity you can summon on a tech product cycle.

In plain terms

Building artificial intelligence takes enormous amounts of electricity, and the power grid simply can't be expanded fast enough — new power lines and the giant transformers they need take years, while a data center goes up in about two. So power, not computer chips, is now the real limit on the AI boom. In the meantime, the extra demand is helping push up everyone's electricity bills, especially near big data centers, which is turning into a political fight. There's a genuine debate about whether the demand is as huge as claimed — some of it may be double-counted — and the buildout actually showed signs of slowing in late 2025.

What you can do now

Expect higher electricity bills and budget for them, especially if you live near data-center growth; check whether your utility has a rate case pending.

Look into time-of-use rates and efficiency. Where utilities offer time-of-use pricing, shifting heavy use (laundry, EV charging) off peak can save money; basic efficiency (LEDs, insulation, a smart thermostat) is the cheapest hedge.

If you invest in AI, remember power is the bottleneck — the picks-and-shovels may be utilities, grid-equipment makers, and power producers as much as chipmakers; diversify rather than betting purely on the AI names.

Engage locally if a data center is proposed near you. Whether ratepayers or the developer pay for grid upgrades is decided in public utility-commission proceedings — those are open to comment.

What to watch

- Grid interconnection queue times and high-voltage transformer lead times — the real speed limit.
- Residential vs data-center electricity rates, and the spread between them — who is actually paying.

- State data-center moratorium bills and ratepayer-protection rules.
- Hyperscaler capex guidance and any further signs of a buildout slowdown (the demand side of the equation).

2. Copper: the metal the electric future can't do without

WHO THIS HITS: *Anyone buying anything electric (cars, appliances, electronics), workers in construction and manufacturing, and investors — copper is in the grid, the EV, and the data center.*

CURRENT THREAT LEVEL: **4/10 — Moderate** *Structural deficit coming, but mostly later*

1	2	3	4	5	6	7	8	9	10

Why this level: A real structural deficit is forming, but near-term the market is roughly balanced to slightly oversupplied, so the acute squeeze is later this decade. **Raises it:** Major mine disruptions stacking up (Grasberg, Kamoa-Kakula); grid + AI + defense demand outpacing supply sooner than expected. **Lowers it:** New supply and recycling ramping on schedule; weak Chinese demand persisting; substitution to aluminium where feasible.

What is happening

[Established] Copper is the physical backbone of electrification — grids, motors, EVs, and data centers are all copper-hungry — and the supply side is struggling to keep up. A January 2026 S&P Global study, *Copper in the Age of AI*, projects demand rising to 42 million metric tons by 2040 (a ~50% increase) while production peaks around 2030 at ~33 million tons, leaving a ~10-million-ton gap — about 25% below demand — which it calls a "systemic risk for global industries." Copper veteran Daniel Yergin frames the quandary neatly: copper is the great enabler of electrification, yet the pace of electrification is itself the challenge for copper.

[Established] Prices already reflect the tension: copper traded above $11,700/ton (a record in December 2025) and topped $12,000 for the first time, after a ~35% gain in 2025. The supply problem is structural — falling ore grades, rising capital costs, permitting delays, water shortages in Chile, and a thin pipeline of new mines — compounded by 2025 disruptions at giant mines (a mud intrusion shut the Grasberg mine; flooding hit Kamoa-Kakula in the DRC).

[Contested] The timing is genuinely debated. The ICSG sees only a small ~150,000-ton deficit in 2026, and Goldman Sachs notes a near-term surplus (with ~1 million tons sitting in US warehouses and Chinese demand soft at −8% year-on-year in Q4 2025), forecasting the real inflection around 2029–2030. So the structural case is widely accepted; the near-term shortage is not yet here.

Why it is under-covered

Copper is boring until it isn't. It lacks the glamour of AI or the drama of oil, and the deficit is dated to the late 2020s, so it stays a back-page commodity story — even though it is the literal wiring of every other story in this set (the grid, the data center, the EV).

The debate

The structural-deficit camp argues: you cannot electrify the world without roughly doubling copper supply, and the mining industry is structurally incapable of delivering it fast enough — so scarcity and high prices are coming.

The near-term-balanced camp argues: right now there is no shortage — inventories are ample and Chinese demand is weak — and high prices plus recycling and substitution (aluminium in some uses) will close more of the gap than the doomers expect.

Both sides accept: ore grades are falling, new mines take 10–20 years to permit and build, and long-run demand from grids, AI, and defense is rising fast.

Stress test (the benign case)

High prices are the cure for high prices: they spur exploration, recycling (which could double), substitution, and efficiency, and a prolonged Chinese slowdown (see Set 1) would soften demand. If supply responds and demand disappoints, the "deficit" arrives later and milder than feared. The vulnerability: the response time is measured in a decade-plus, ore grades keep falling regardless of price, and a cluster of mine disruptions can turn a forecast surplus into a real shortage in a single year.

Historical parallel

The 2000s commodity supercycle is the template — Chinese industrialization drove copper from under $2,000 to over $10,000 a ton before supply caught up. The recurring pattern: structural demand shifts arrive faster than mines can be built, prices overshoot, capital floods in, and supply eventually responds — but the lag is long, and the interim is volatile and expensive.

In plain terms

Electric everything — power grids, electric cars, data centers — needs a lot of copper, and the world is on track to need far more than mines can dig up. New mines take 10 to 20 years to open, ore is getting lower quality, and big mines keep hitting problems. Experts call the looming gap a "systemic risk." The catch: the real shortage is mostly expected later this decade (around 2029–2030); right now there's actually plenty of copper and prices, while high, reflect the future more than the present. It's a slow-moving squeeze, not an emergency — but it's the wiring under almost every other story in this report.

What you can do now

Mostly awareness: expect electrification (EVs, grid upgrades, appliances) to stay expensive partly because the raw metal is getting pricier over time.

If you invest in the energy transition or AI, copper and other 'picks-and-shovels' materials are an under-appreciated way to play it — but commodities are volatile, so size positions modestly and expect a bumpy ride, not a straight line.

Don't chase the copper trade on a headline. The structural story is real but the near-term market is well-supplied; buying after a price spike is how people get hurt in commodities.

Recycle your electronics and metals. Recycled ('secondary') copper is a genuine part of the solution, and e-waste recycling keeps valuable metal in the system.

What to watch

- The copper price and exchange inventories (LME/COMEX/Shanghai) — falling stocks signal the deficit arriving.
- Major mine disruptions and production guidance (Chile, Peru, DRC, Indonesia).
- Chinese demand and stimulus — the swing factor for near-term balance.
- Mine permitting and new-project sanctioning — the only real cure for the long-run gap.

3. Water: the economy’s most underpriced input

WHO THIS HITS: *Everyone who eats, and anyone near agriculture, chip plants, or data centers — water is the hidden input in food, chips, and the cloud, and it’s priced as if it were free.*

CURRENT THREAT LEVEL: **4/10 — Moderate** *Slow and underpriced, but regionally acute*

1	2	3	4	5	6	7	8	9	10

Why this level: A slow, structurally underpriced constraint that is already regionally acute (chips, farms, data centers) but lacks a single near-term systemic trigger. **Raises it:** A drought hitting a chip hub (Taiwan, Arizona) or a major breadbasket; escalation of a transboundary water dispute (Nile, Indus, Colorado). **Lowers it:** Wetter cycles easing key basins; efficiency, reuse, and — above all — sane water pricing scaling up.

What is happening

[Established] Roughly half the world’s population already faces water scarcity, and the Global Commission on the Economics of Water warns that, on current trends, water stress could cut global GDP by around 8% by 2050 (and far more in lower-income countries), with over half of food production at risk. Agriculture uses about 70% of the world’s freshwater, so water and food are the same story. The Commission’s core economic point is blunt: water is systematically *underpriced*, which encourages waste and pushes water-hungry industries into the wrong places.

[Established] The collision with the tech boom is direct. An analysis by the TNFD found ~40% of existing chip fabs sit in basins projected to face high or extreme water stress by 2030. Taiwan — where TSMC makes ~90% of the world’s most advanced chips — suffered its worst drought since 1964 in 2020–21, cutting irrigation to ~74,000 hectares of rice so the fabs could keep running: water for chips over water for food, made explicit. Data centers add thirst on top: community opposition over water has delayed or blocked tens of billions of dollars of projects.

[Established] Transboundary disputes raise the stakes. The Colorado River — underpinning ~$1.4tn of US economic activity — has run a structural deficit for over 20 years. Egypt depends on the Nile for ~97% of its water; its per-capita availability has fallen below the UN’s scarcity threshold, and filling Ethiopia’s Grand Renaissance Dam could sharply cut downstream supply.

Why it is under-covered

Because water is nearly free, it doesn’t show up as a market signal until it’s a crisis. There is no global “water price” on a screen, so the slow erosion of this input is invisible to markets — right up until a drought stops a chip fab, empties a reservoir, or sparks a cross-border dispute.

The debate

The crisis camp argues: the global water cycle is being pushed out of balance, water is dangerously mispriced, and the result is a slow-building hit to food, industry, and growth — a financial risk markets are ignoring.

The techno-optimist camp argues: efficiency, recycling/reuse, desalination, precision irrigation, and (eventually) better pricing can stretch supply enormously, and the largest firms can secure their own water — so it’s a manageable engineering and policy problem, not a catastrophe.

Both sides accept: water is underpriced, agriculture dominates use, and water-intensive industries are increasingly sited in already-stressed basins.

Stress test (the benign case)

Water problems are mostly local and solvable: efficiency and reuse have huge headroom, chipmakers already recycle most process water, and pricing reform (where politically possible) would reallocate water quickly. Most of the world is not at the extreme tail. The vulnerability is that water is intensely political and local: you cannot ship it far or cheaply, drought is getting more frequent, and a single dry year in the wrong basin — Taiwan, Arizona, the Nile — can disrupt chips, food, or geopolitics with little warning.

Historical parallel

History is full of water-driven economic and political upheaval, from the US Dust Bowl of the 1930s to Cape Town's 2018 "Day Zero" near-miss and Chennai running dry in 2019. The recurring lesson: societies treat water as infinite until a sharp shortage forces brutal rationing and reveals how much of the economy was silently resting on it.

In plain terms

Water is the invisible ingredient in food, computer chips, and data centers — and because it's priced as though it were free, we waste it and build thirsty industries in dry places. Farming uses about 70% of fresh water, so water shortages are really food shortages. The tech boom collides with this directly: Taiwan, which makes most advanced chips, once cut water to rice farms during a drought so the chip factories could keep running. None of this is a single global emergency — it's slow and very local — but a bad drought in the wrong place (a chip hub, a major farming region, a shared river) can cause outsized economic damage quickly.

What you can do now

This is mostly awareness — but understanding that food prices, chip supply, and some regional economies hinge on water helps you read the news (a drought story can be an economic story).

If you live in a water-stressed region, water-efficient appliances, fixing leaks, and drought-tolerant landscaping cut bills and exposure — and water rates are likely to rise over time as scarcity gets priced in.

Factor water into big location decisions. If you're buying property or a business, a region's long-term water situation is a real and under-considered risk.

For investors, water (utilities, efficiency, treatment) is a long-duration theme, and water stress is a genuine risk hidden inside agriculture, semiconductors, and data-center holdings — worth checking, not panicking over.

What to watch

- Drought conditions in key chip and farming regions (Taiwan, US Southwest, India, Brazil).
- Reservoir levels on stressed systems (Colorado River/Lake Mead) and major dam disputes (Nile/GERD, Indus).
- Moves toward real water pricing, water markets, and reuse mandates — the structural fix.
- Data-center and fab approvals being blocked or relocated over water — the economic bite.

4. Food and fertilizer: a concentrated, fragile system

WHO THIS HITS: *Everyone who eats — especially lower-income households and import-dependent countries — and farmers facing record input costs. Fertilizer shocks become food-price shocks within a year.*

CURRENT THREAT LEVEL: **6/10 — Elevated** *Active export curbs and Gulf-input risk*

1	2	3	4	5	6	7	8	9	10

Why this level: Active export curbs (China through August, Russia controls), Strait of Hormuz disruption to fertilizer inputs, and an FAO warning that staple-crop output could fall within 6–9 months keep this elevated. **Raises it:** A prolonged Hormuz/Red Sea disruption to sulfur and ammonia; a poor harvest; new export bans cascading across producers. **Lowers it:** Export curbs lifting on schedule; new fertilizer capacity arriving in 2026–27; a calm Gulf and a good growing season.

What is happening

[Established] The world's food supply rests on a handful of fertilizer producers and trade routes, and several are constrained at once. China — a top phosphate and nitrogen exporter — has restricted fertilizer exports (quotas and licensing) into late August 2026 to protect domestic supply (its 2024 nitrogen exports fell more than 90% year-on-year). Russia, a dominant nitrogen exporter, suspended export licenses for ammonium nitrate; Belarusian potash remains under sanctions; and the EU has tariffed Russian and Belarusian fertilizer. Prices sit roughly 17–20% above year-ago levels.

[Established] The 2026 Gulf conflict makes it worse. Large volumes of urea, ammonia, phosphates, and sulfur — critical fertilizer inputs — move through the Strait of Hormuz, and the FAO warns that reduced availability could lower output of wheat, maize, and rice within 6–9 months and push food prices up. Countries are scrambling for workarounds: India locked in Canadian and rerouted Russian supply; the US expanded sanctions waivers for Venezuelan fertilizer.

[Established] Underneath the headlines sit two structural concentrations. Canada supplies roughly 90% of US potash imports — one analyst's blunt summary: the US "cannot feed itself… without either Canada or Russia." And phosphate is doubly squeezed: China is diverting it into lithium-iron-phosphate EV batteries (a direct link to electrification), while Morocco holds the lion's share of the world's phosphate-rock reserves. [Contested] Estimates that global phosphorus production could peak around 2035 are debated, but the concentration of reserves is not.

Why it is under-covered

Fertilizer is invisible until food gets expensive. It's a chemistry-and-logistics story most people never think about — yet roughly half the world's food production depends on synthetic fertilizer, and a fertilizer shock today is a grocery-bill and hunger shock 6–12 months later, especially in poorer importing countries (a direct link to the Set 1 developing-debt story).

The debate

The fragility camp argues: production and trade are dangerously concentrated in a few countries and chokepoints, export bans are spreading, and the Gulf conflict plus phosphate scarcity set up a multi-year food-security risk.

The reassurance camp argues: fertilizer markets are adapting (rerouting, new capacity coming in 2026–27), affordability relative to food prices is still near pre-2020 norms, and bans tend to be temporary — so this is volatility, not collapse.

Both sides accept: supply is concentrated, prices are elevated, key inputs run through contested waters, and the lag from fertilizer to food means today's disruptions show up in next year's harvests.

Stress test (the benign case)

The system has proven adaptable: after the 2022 shock, trade rerouted and prices eventually fell. New capacity (potash in Canada, phosphate elsewhere) is arriving, export bans are usually lifted within a season, and high prices pull supply forward. The vulnerability is correlation and timing: if a prolonged Gulf disruption hits sulfur/ammonia during planting season while China's curbs persist and a major harvest disappoints, the buffers thin fast — and the people hurt first are those who can least afford it.

Historical parallel

2022 is the fresh memory: Russia's invasion of Ukraine spiked fertilizer and grain prices, drove food inflation worldwide, and contributed to unrest in import-dependent countries. Further back, the 2007–08 food-price crisis — amplified by export bans — triggered riots across dozens of countries. The pattern: fertilizer and grain shocks transmit into political instability with a lag, and export bans make a shortage worse by turning it into a panic.

In plain terms

Most of the world's food depends on fertilizer, and fertilizer comes from just a few countries and travels through a few sea routes — several of which are now restricted or contested at once. China is limiting exports into late summer, Russia has curbed some too, and key fertilizer ingredients pass through the Strait of Hormuz, which the Gulf conflict has disrupted. The UN warns this could lower wheat, corn, and rice harvests within 6–9 months and raise food prices. There's also a deeper vulnerability: the US gets about 90% of one key fertilizer (potash) from Canada, and the world's phosphate is heavily concentrated in a few places — with China now diverting some into EV batteries. It tends to be volatility rather than collapse, but it hits grocery bills, and the poorest countries first.

What you can do now

Expect food-price volatility over the next year and build a little flexibility into your grocery budget; staples (grains, cooking oil) are the most exposed.

A modest food buffer is reasonable, panic-buying is not. A couple of weeks of shelf-stable staples is sensible household resilience; hoarding makes shortages worse for everyone (that's exactly how the 2008 crisis spiraled).

Reduce food waste. Roughly a third of food is wasted; cutting your own waste is the most direct way to blunt higher prices and ease pressure on the system.

If you garden or can buy local, even a small amount of home or local food production adds personal resilience to global supply shocks.

What to watch

- Fertilizer prices (urea, DAP, potash) and the FAO Food Price Index — the leading and lagging gauges.
- China's export-curb expiry (late August 2026) and any Russian/other new restrictions.

- Strait of Hormuz / Red Sea status as it affects sulfur and ammonia flows.
- Harvest and planting reports in major exporters (US, Brazil, Black Sea, India) — where shortages become real.

5. The lithography chokepoint: one machine the whole digital world needs

WHO THIS HITS: *Everyone who uses any modern technology, and anyone invested in AI or chips — every advanced chip on Earth depends on one company's machines, made with one supplier's optics, in one country.*

CURRENT THREAT LEVEL: **5/10 — Elevated** *Extreme single point of failure*

1	2	3	4	5	6	7	8	9	10

Why this level: The most concentrated chokepoint in the entire economy — one company, one optics supplier, key fabs in Taiwan — though there is no imminent disruption. **Raises it:** A Taiwan crisis or an incident at ASML (Veldhoven) or Zeiss; a sharp escalation of export-control retaliation. **Lowers it:** Geographic diversification of advanced fabs (US, Japan, Europe) maturing; stable cross-strait and export-control conditions.

What is happening

[Established] Every advanced chip in the world — every AI accelerator, every flagship phone processor — is made using extreme-ultraviolet (EUV) lithography machines built by exactly one company: ASML, in the Netherlands. ASML has a 100% monopoly on EUV and ~94% of lithography overall; Nikon and Canon exited advanced lithography years ago. ASML can build only ~50–60 of these machines a year — a physical ceiling on how much leading-edge chip capacity the world can add. Each next-generation High-NA machine costs around $400 million.

[Established] The concentration runs deeper than one company. ASML itself depends on a single optics supplier — Germany's Zeiss — for the flawless mirrors EUV requires; the future pace of the technology is effectively gated by what Zeiss can build. The leading-edge chips themselves are made overwhelmingly by TSMC in Taiwan (which also depends on Taiwan's water, per Story 3). So the entire digital economy funnels through one machine-maker, one optics-maker, and one island.

[Established] This chokepoint is also a weapon. EUV machines have never been sold to China, and US-led export controls keep tightening; China's share of ASML sales fell from ~33% in 2025 toward ~19–20% in 2026. China is reportedly a decade or more behind on domestic EUV. **[Contested]** In April 2026 TSMC said it would stick with current-generation EUV through at least 2029 rather than buy High-NA immediately — a reminder that even the indispensable supplier has only one buyer big enough to absorb its bleeding edge, a mutual dependence that cuts both ways.

Why it is under-covered

ASML is the most strategically important company almost no one outside the industry can name. The chokepoint is abstract (a lithography machine in Veldhoven) and has been stable for years, so it hides in plain sight — until a Taiwan scenario or a single-site incident makes its fragility suddenly concrete.

The debate

The fragility camp argues: this is the ultimate single point of failure — one firm, one optics supplier, one island — and a disruption (war over Taiwan, an accident, escalating controls) would halt advanced-chip progress for the whole world.

The resilience camp argues: the very irreplaceability creates caution on all sides (no one benefits from breaking it), diversification is underway (new fabs in the US, Japan, and Europe), and older chips can carry much of the economy for a long time.

Both sides accept: the concentration is extreme and real, processing/manufacturing know-how takes decades to replicate, and meaningful diversification is years away.

Stress test (the benign case)

Mutual dependence is stabilizing: ASML needs TSMC as much as TSMC needs ASML, China is deterred from a move on Taiwan partly because it would destroy the thing it wants, and a multi-year push is spreading advanced fabs across the US, Japan, and Europe. Older-node chips — which run most of the physical economy — are far less concentrated. The vulnerability: none of that helps in the tail event. A single bad day in the Taiwan Strait, in Veldhoven, or at Zeiss has no quick substitute, and "years away" diversification is no comfort on the day it matters.

Historical parallel

The 2020–22 chip shortage — when a shortfall of even cheap, mature chips idled auto plants worldwide — was a preview of how a narrow bottleneck cascades through the real economy. The deeper analogue is the oil chokepoints of the 1970s: a single concentrated input, once disrupted, reorders the global economy and triggers a scramble for alternatives that takes a decade to bear fruit.

In plain terms

Every cutting-edge computer chip — in AI systems, phones, everything — is made using a machine that only one company on Earth knows how to build: ASML, in the Netherlands. That company in turn depends on a single supplier (Zeiss in Germany) for its precision mirrors, and the chips themselves are mostly made by one company in Taiwan. So the entire digital world runs through one machine-maker, one optics-maker, and one island. It's been stable for years, and everyone has reasons not to break it — but it's the most concentrated single point of failure in the global economy, and a war over Taiwan or a serious accident would have no quick fix.

What you can do now

This is awareness, not an action item — but it explains why 'Taiwan risk' is treated as a global-economy risk, not a regional one.

Mind concentration in your investments. AI and chip exposure ultimately funnels through a very small number of irreplaceable companies and one geography; broad diversification (across regions and sectors) is the realistic hedge for an ordinary investor.

Expect tech-hardware price and availability swings whenever export controls or cross-strait tensions escalate; big-ticket electronics purchases can be worth timing around obvious flashpoints.

Don't try to trade the geopolitics. Tail-risk timing is a fool's errand; a cash buffer and diversification protect you better than predicting a Taiwan crisis.

What to watch

- Cross-strait tensions and any military escalation around Taiwan — the dominant tail risk.
- Export-control rounds and Chinese retaliation (and China's share of ASML sales).
- Progress and yields at new advanced fabs outside Taiwan (US, Japan, Europe) — real diversification vs announcements.
- ASML order backlog and High-NA adoption timelines — the health of the chokepoint itself.

6. Sea lanes and undersea cables: the physical arteries of globalization

WHO THIS HITS: *Everyone — through fuel and goods prices, and through the internet itself. About 80% of trade moves by sea and ~95% of data moves under it, much of it through a few narrow, contested passages.*

CURRENT THREAT LEVEL: 7/10 — High *A live energy-and-trade crisis*

1	2	3	4	5	6	7	8	9	10

Why this level: A live crisis: the Strait of Hormuz was closed for months, the Red Sea remains contested, undersea cables are openly threatened, and the IEA calls the resulting energy shock worse than the 1970s. **Raises it:** A breakdown of the Hormuz ceasefire; a major cable-cutting event; a second chokepoint (Bab el-Mandeb, Malacca, Panama) seizing at the same time. **Lowers it:** A durable Gulf ceasefire reopening Hormuz; restored Red Sea transit; cable redundancy and protection improving.

What is happening

[Established] Globalization runs through a handful of narrow passages, and several are under stress at once. The Strait of Hormuz — through which roughly a fifth of the world's traded oil moves — was effectively closed for months in 2026 amid the US–Iran conflict, stranding up to ~20,000 seafarers on ~2,000 vessels; the IEA called the resulting oil-and-gas shock more severe than the crises of 1973, 1979, and 2022 combined. A US–Iran ceasefire is now slowly reopening it, but the situation is fragile, and on 8 June 2026 the Houthis declared a renewed ban on Israeli ships in the Red Sea.

[Established] The same chokepoints are also **digital** chokepoints. Around 95% of intercontinental data travels through undersea cables, and they cluster through the same narrow waters: roughly 90% of Europe–Asia cables run through the Red Sea, and multiple major systems thread the Strait of Hormuz. In 2024, cable cuts in the Red Sea (one caused by a ship's dragged anchor) disrupted an estimated 25% of Europe–Asia internet traffic. In 2026, Iran-linked media pointedly highlighted the Hormuz cables' vulnerability, and Meta paused work on a major Gulf cable system. Unlike ships, which can reroute over weeks, cable outages hit financial trading, cloud services, and communications within hours — and repairs can take weeks to months.

[Established] Other chokepoints are flashing too. The Panama Canal cut transits ~40% during the 2023–24 drought (a water link to Story 3), and in 2026 became a US–China flashpoint after Panama's top court voided a Hong Kong company's port contracts — prompting China to detain scores of Panama-flagged vessels. The Strait of Malacca (80,000+ vessels a year) saw Indonesia float a transit toll. The deeper shift: disruption no longer needs a formal blockade — drones, mines, cyberattacks, dragged anchors, and insurance spikes can achieve the same effect far more cheaply.

Why it is under-covered

The oil-tanker angle gets covered; the cable angle almost never does, even though a coordinated cable disruption could hit the economy as hard as closing a canal. These are the literal arteries of trade and data, owned mostly by private firms, governed by weak international frameworks, and invisible until they're cut.

The debate

The fragility camp argues: the world rebuilt itself around just-in-time, open sea lanes and a few cable corridors, and that assumption is now broken — cheap, deniable attacks can choke trade, energy, and data at multiple points at once.

The resilience camp argues: shipping reroutes (around the Cape, as it did for the Red Sea), cables get repaired and rerouted, and markets adjust — painfully but without collapse — as the post-2023 Red Sea diversions showed.

Both sides accept: trade and data are concentrated through a handful of chokepoints, redundancy and protection are thin, and the cost of disrupting them has fallen dramatically.

Stress test (the benign case)

The system is more adaptable than it looks: ships rerouted around the Cape of Good Hope after 2023 (slower and costlier, but it worked), cables can be repaired and new routes built, and a Gulf ceasefire could reopen Hormuz and calm the worst of it. The vulnerability is simultaneity and speed: rerouting buys time only if one chokepoint fails at a time, and cable cuts give no warning. A coordinated or compounding event — Hormuz plus the Red Sea plus a cable cut — would overwhelm the slack the system relies on.

Historical parallel

The 2021 grounding of a single container ship in the Suez Canal — which blocked ~12% of global trade for six days and snarled supply chains for months — showed how one chokepoint, briefly closed, ripples worldwide. The 1956 and 1967 Suez closures reordered global shipping for years. The lesson: chokepoints concentrate efficiency in good times and catastrophe in bad ones, and the world keeps rediscovering this the hard way.

In plain terms

Almost everything you buy travels by sea, and almost all internet data travels through cables on the ocean floor — and both funnel through a few narrow passages like the Strait of Hormuz and the Red Sea. Right now several of these are in trouble at once: the Gulf conflict shut the Strait of Hormuz for months (causing an energy shock the IEA calls worse than the 1970s), the Red Sea is contested, and the undersea cables running through these same waters are being openly threatened. Ships can slowly reroute the long way around; cables can't — cutting them slows the internet, trading, and cloud services within hours. It's the most active crisis in this report, easing a little as a ceasefire reopens Hormuz, but fragile.

What you can do now

Expect fuel and goods prices to swing with Gulf and Red Sea headlines; energy-shock inflation can show up at the pump and in shipping-dependent goods within weeks.

Keep a basic resilience kit. Because outages (including internet/communications from cable problems) are possible, the standard advice applies: some cash on hand, a little extra of essential medications and supplies, and offline copies of critical documents.

Don't over-react in markets to each headline. Energy and shipping shocks are real but often partly priced in; a cash buffer beats panic-trading the geopolitics.

For businesses, this is the case for mapping your supply chain's chokepoint exposure and holding sensible buffer inventory — just-in-time has a hidden cost when arteries clog.

What to watch

- The Strait of Hormuz ceasefire and reopening — and any breakdown; oil and tanker-insurance rates.
- Red Sea / Bab el-Mandeb transit volumes and any new cable-cutting incidents.
- Panama Canal water levels and the CK Hutchison port dispute; Malacca governance moves.
- Shipping rates (container and tanker) and marine insurance premiums — the price of chokepoint risk.

7. Clearinghouses: the financial system's hidden single point of failure

WHO THIS HITS: *Everyone with savings in markets — a handful of clearinghouses sit at the center of the financial system, and one failing would be a system-wide event.*

CURRENT THREAT LEVEL: **4/10 — Moderate** *Concentrating risk, no near-term trigger*

1	2	3	4	5	6	7	8	9	10

Why this level: Clearinghouses concentrate the system's residual risk in a few 'too-important-to-fail' hubs, and the US Treasury-clearing mandate will push even more through them — but failures are historically rare and buffers are large. **Raises it:** A volatility spike forcing huge procyclical margin calls (cf. UK 2022 LDI, nickel 2022); a large clearing-member default. **Lowers it:** Smooth Treasury-clearing implementation with robust capacity; stronger CCP recovery and resolution plans; calmer markets.

What is happening

[Established] After 2008, regulators pushed vast swaths of trading through central counterparties (CCPs) — clearinghouses that stand between buyer and seller to guarantee trades. This genuinely reduced the tangled, bilateral counterparty risk that amplified the crisis. But it also concentrated the system's residual risk into a handful of "too-important-to-fail" hubs — names like CME, ICE, LCH, Eurex, and DTCC/FICC. A Chicago Fed analysis finds the clearing ecosystem is now *more concentrated than it was in 2007*, before the crisis.

[Established] The mechanism that makes CCPs safe in normal times can make them dangerous in a panic. CCPs require members to post cash margin that rises automatically with volatility — so their liquidity demands are inherently *procyclical*: exactly when markets are stressed and cash is scarce, the clearinghouse calls for more of it, which can force fire-sales and propagate the stress it was meant to contain. The Fed has flagged this systemic liquidity risk explicitly.

[Established] The live development: the SEC's mandate to centrally clear US Treasuries (cash by end-2026, repo by mid-2027) — the regulators' fix for the basis-trade risk in Set 1 — will route an enormous additional volume through CCPs, *increasing* concentration. In other words, the cure for one chokepoint (hidden leverage in Treasuries) deepens another (clearinghouse concentration).

Why it is under-covered

CCPs are deliberately invisible plumbing — the system works precisely because no one thinks about them. There's no retail hook and the machinery is arcane (default waterfalls, margin models, mutualized loss funds). Only three CCPs have ever failed (in 1974, 1983, and 1987), which breeds a complacency that the consequences of a fourth would shatter.

The debate

The reassurance camp argues: CCPs work — they survived 2008, 2020, and 2022 — with deep, pre-funded default buffers, strict margining, and far more transparency than the bilateral web they replaced; concentration brings standardization and oversight.

The concern camp argues: concentration plus procyclical margin calls turns CCPs into potential transmission mechanisms; their risk is poorly understood; recovery and resolution plans are untested; and the Treasury mandate is loading still more onto them.

Both sides accept: central clearing reduced bilateral counterparty risk but concentrated residual risk, margin demands are procyclical, and the few key CCPs are now critical infrastructure.

Stress test (the benign case)

The buffers are real and large: pre-funded default funds, layered "waterfalls," conservative margining, and stress tests designed around multiple simultaneous member defaults. CCPs absorbed the 2020 dash-for-cash and the 2022 commodity and gilt shocks without failing. The vulnerability is the tail no buffer is sized for: a fast, correlated volatility spike that triggers margin calls system-wide (as the UK's 2022 LDI episode and the 2022 nickel-market chaos foreshadowed), or the failure of a large clearing member — against a backdrop of rising concentration.

Historical parallel

The clearest warnings are recent and partial: the UK's September 2022 pension (LDI) crisis, where margin calls on leveraged hedges forced a doom-loop of gilt selling until the Bank of England intervened; and the March 2020 "dash for cash," when margin demands drained liquidity system-wide. The 1987 crash nearly broke the clearing system and did topple the Hong Kong Futures Exchange's clearer. The pattern: clearing is invisible and reassuring until a volatility spike turns its safety mechanism into an accelerant.

In plain terms

After the 2008 crisis, regulators routed most trading through a few 'clearinghouses' — middlemen that sit between buyers and sellers and guarantee trades. This made the system safer in normal times, but it piled the leftover risk into a handful of giant hubs that are now 'too important to fail.' The catch: these hubs demand more cash from their members exactly when markets are panicking and cash is scarce — which can make a crisis worse. And a new rule pushing US government-bond trading through them (the fix for a problem in Set 1) is concentrating even more risk in one place. Failures are very rare and the safety buffers are big, so this is a low-probability tail risk — but if it ever broke, it would be a system-wide event.

What you can do now

This is awareness — there's no direct retail action, but it's part of why broad diversification and a cash buffer matter: in a true plumbing crisis, everything sells off together for a while.

Keep an emergency fund outside the markets. The recurring lesson of every plumbing scare (2008, 2020, 2022) is that cash on hand means you're never forced to sell at the bottom.

Be skeptical of leverage and 'can't-lose' hedges. The 2022 UK pension blow-up came from leveraged strategies that needed cash exactly when it vanished; the same logic applies to your own finances — avoid being a forced seller.

Don't lose sleep over this one specifically. It's a low-probability tail risk handled by people whose job is to handle it; the general 'cash buffer + diversify' habit is the appropriate response.

What to watch

- Market volatility (VIX/MOVE) and margin-call episodes — the trigger for procyclical stress.
- US Treasury-clearing implementation (end-2026 cash, mid-2027 repo) and whether clearing capacity scales smoothly.
- CCP stress-test results and recovery/resolution planning from regulators.

- Any large clearing-member distress — the most direct failure path.

Pattern analysis: how the chokepoints interlock — and link to Sets 1 & 2

These seven are not separate. They are one chain, and they connect directly to the financial stories in Sets 1 and 2.

The chain runs through itself

Follow the AI boom downward and it passes through every story in this set: AI needs power (1), power and AI need copper (2), copper mining and chip-making need water (3), water competes with food and fertilizer (4), the chips themselves come from one machine and one island (5), everything physical moves through contested sea lanes while the data moves through the cables beneath them (6) — and the whole edifice is financed and cleared through a few hubs (7). Pull any link and the others feel it. The 2026 Gulf conflict is the clearest example: it hits energy, fertilizer inputs, sea lanes, and cables simultaneously.

Theme A — The abstract economy has a physical body

The defining illusion of the digital age is that growth is weightless — software, services, AI, finance. Set 3 is the rebuttal. Every "cloud" is a building drawing megawatts; every model runs on metal dug from a falling-grade ore body; every chip is etched by one machine and cooled by water that a farmer also needs. The economy didn't dematerialize. It just stopped looking at its own foundations.

Theme B — Decades of treating vital inputs as free and infinite

Each chokepoint is the bill for a long holiday from physical reality. Power, water, copper, fertilizer, open sea lanes — all were treated as abundant and cheap, so we under-invested in them and over-built on top of them. Water is the purest case: it is literally underpriced, so it is wasted and misallocated. The pattern is the mirror image of Set 2's "extend and pretend" — there, the financial system postponed recognizing losses; here, the real economy postponed recognizing limits.

Theme C — Concentration is efficient until it's catastrophic

Every story is a single point of failure: one machine-maker, a few mines and fertilizer exporters, a few sea lanes and cable corridors, a few clearinghouses. Concentration is wildly efficient in calm times — it's why the system evolved this way — and wildly dangerous in a shock. This is the same shape as Set 1's hidden leverage and Set 2's index concentration: the system optimizes for normal and is fragile to the tail.

The link back to Sets 1 & 2

The connections are concrete, not thematic. The AI power and copper bottlenecks (1–2) are the physical constraint on the AI-capex boom from Set 2 — the reason that debt-fueled spending might not earn out. The clearinghouse story (7) is the direct sequel to Set 1's basis trade: the regulators' fix (mandatory Treasury clearing) solves hidden leverage by concentrating risk in CCPs. Fertilizer and food (4) land hardest on the developing-world debtors of Set 1. And the Gulf energy shock that runs through Set 3's sea lanes is the same shock pressuring the inflation, central-bank, and sovereign-debt threads in Sets 1 and 2. It is all one system.

In plain terms (the whole picture)

The booming digital and financial economy — AI, the cloud, the markets — sits on top of a physical foundation we mostly stopped paying attention to: electricity, copper, water, food and fertilizer, the single machine that makes advanced chips, the sea lanes and undersea cables that carry trade and data, and the few financial hubs that clear it all. For decades we treated these as free and infinite, under-invested in them, and built more and more on top. Now several are straining at once — and they're all connected, so a squeeze in one (say, the Gulf conflict choking shipping) ripples into the others (energy, fertilizer, prices). None of this is doom; most are slow-moving and fixable with investment and better pricing. But the takeaway is the same humble one as before: the 'weightless' economy has a very physical body, and the smart personal response is steady — keep a cash and supplies buffer, diversify, avoid leverage, and don't assume the cheap, reliable foundation will always be there.

Master watchlist

The highest-signal indicators across all seven stories, with realistic update frequency.

Indicator	Why it matters / what it signals	Cadence
Grid interconnection & transformer lead times	The real speed limit on AI power — and on the AI-capex payoff.	Quarterly
Residential vs data-center electricity rates	Who pays for the buildout; the political pressure gauge.	Monthly
Copper price + exchange inventories	Falling stocks signal the structural deficit arriving.	Daily
Drought in chip/farm hubs; key reservoir levels	Where water stress turns into economic disruption.	Ongoing
Fertilizer prices + FAO Food Price Index	Fertilizer shocks become food-price shocks within ~6–12 months.	Monthly
Hormuz/Red Sea status; tanker & container rates	The live chokepoint crisis — energy, trade, and cables.	Daily
Undersea cable incidents	Instant hits to data, finance, and cloud services.	Event
Market volatility (VIX/MOVE) + Treasury-clearing rollout	The trigger and the loading of clearinghouse concentration risk.	Daily / Milestone

Plain-language glossary

Quick, jargon-free definitions for the terms used above.

Grid interconnection queue — the waiting line to connect a new power plant (or big user) to the electricity grid; now multi-year in the US.

Transformer / grid equipment — the heavy hardware that moves and steps electricity up and down; high-voltage transformers now have 2–4-year lead times.

Behind-the-meter / on-site power — generating your own electricity at a site (e.g. on-site gas or solar) instead of drawing it from the public grid.

Gigawatt (GW) / terawatt-hour (TWh) — units of power and energy; a large data center can need ~1 GW, roughly the output of a big power plant.

Capacity market — payments to power plants to be available when needed; data-center demand sent some of these prices up sharply.

Structural deficit — a lasting shortfall where demand persistently outruns supply — not a temporary blip.

Ore grade — how much metal is in the rock; falling copper grades mean more digging and cost for the same metal.

EUV lithography — extreme-ultraviolet light used to print the tiniest chip circuits; only ASML makes the machines.

High-NA EUV — the next, more powerful (and ~$400m) generation of EUV machines for the smallest chips.

Fab / foundry — a semiconductor fabrication plant; TSMC in Taiwan makes most of the world's most advanced chips.

Chokepoint — a narrow, hard-to-replace passage — physical (a strait) or structural (one supplier) — that lots of activity must pass through.

Strait of Hormuz — the narrow Gulf passage carrying ~20% of traded oil; effectively closed for months in 2026.

Bab el-Mandeb / Red Sea — the route linking the Suez Canal to the Indian Ocean; carries major trade and ~90% of Europe–Asia data cables.

Undersea (submarine) cable — fiber-optic cable on the ocean floor; carries ~95% of intercontinental internet data.

Potash / phosphate / urea — the three main fertilizer types (potassium, phosphorus, nitrogen) that underpin most food production.

DAP / MOP — common traded fertilizers: diammonium phosphate and muriate of potash.

Central counterparty (CCP) / clearinghouse — a financial middleman that guarantees trades; concentrates risk in a few hubs.

Margin call — a demand to post more cash as collateral when markets move; rises automatically when volatility spikes ('procyclical').

Default waterfall — the layered pile of buffers (margin, then default fund, then mutualized losses) a clearinghouse uses if a member fails.

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End of Set 3. This set was the physical foundation under Sets 1 and 2. Per our earlier discussion, the clearly-strong material is starting to thin — the next set would likely cover money-market funds and the repo system, the offshore-dollar (eurodollar) market, semiconductor materials beyond lithography, and demographic collapse in specific countries (Korea, Italy) — strong topics, but several rhyme with mechanisms we've already covered. A good moment to consider shifting to periodic updates (via the tracker) plus a smaller 'what's newly emerged' set, rather than chasing volume. Your call.